

cant customer to them.

There are many myths surrounding analogue ASICs that prevent OEMs from exploring opportunities to lower costs, reduce size and protect IP. Let us look at some of these.

Myth #1. It is economical to integrate analogue functions into an ASIC only if analogue content is minimal.

FALSE. Actual manufacturing cost of an analogue ASIC chip can provide huge savings when compared to the collective costs of the ICs it replaces. However, there are other costs associated with the ASIC that must be considered and amortised over the life of the product.

Non-recurring engineering costs, based on the complexity of the design as well as hand tooling costs such as masks and test hardware, can add cost to the ASIC chip cost, depending on complexity and lifetime volume of the device. Some analogue ASIC companies offer rebates to these fixed costs, making these even more cost effective. Be sure to ask for a rebate when getting a quote.

It is as important to know what to incorporate into the ASIC as it is to know what should remain a discrete component.

Most analogue applications use a collection of passive elements and discrete transistors in addition to the ICs involved. Integrating as many of these components as possible to the ASIC often comes for free and can have a dramatic effect in lowering the end product's total assembly cost as well as size and cost. It is this potential total system cost saving that bolsters the justification to develop the analogue ASIC.

Myth #2. Mixed-signal ASIC means the same thing as analogue ASIC.

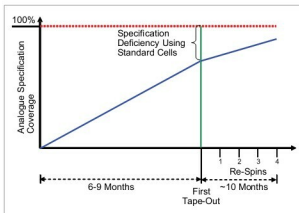


Fig. 2: Graph showing typical inherent errors in achieving desired analogue specification when using standard mixed-signal cells

FALSE. The term mixed-signal implies a combination of analogue and digital circuitry on a single chip. Companies that focus on mixed-signal typically use cell libraries for their analogue needs. On the other hand, companies that make analogue ASICs do not use cell libraries. Rather, they handcraft each design to exactly meet the required specifications.

There is a distinct difference in the engineering skill levels required to combine library cells on a silicon chip versus actually creating a handcrafted analogue design that uniquely satisfies all requirements of the specification. For some simple applications, analogue library cells offer sufficient performance to meet the system requirements.

However, increased sophistication of analogue applications necessitates designs that are truly application-specific and not a compilation of general-purpose analogue cell blocks. When using cell blocks, designers are forced to make compromises by picking the cell (whether amplifier, reference, PLL, etc) that most closely meets the application's need. Compromises accumulate and soon the end design is unable to meet its original specification.

Like the big analogue IC companies, analogue ASIC companies employ experienced analogue designers who are artisans at analogue

invention. Many of them have spent decades at the big analogue companies (like National, LT, Maxim and TI), learning from the industry gurus.

Be careful not to let a mixed-signal design house negotiate you away from your ideal specification. Close is not good enough; analogue must be exact!

Myth #3. Only ultra-high-volume applications can benefit from analogue ASICs.

FALSE. As noted earlier, many large semiconductor companies focus their ASIC efforts onto a handful of very large customers. All full-service ASIC houses have their own business criteria regarding minimum non-recurring engineering (NRE) costs, tooling and, most importantly, annual volume. Some ASIC houses avoid the issue by just offering design services and leaving the issue of manufacturing to the customer. This is a bad choice for you, unless you have IC manufacturing experts on your team.

Throughout the world and in particular in Silicon Valley, California, USA, there are numerous boutique wafer fabs that specialise in analogue processes and accept lower volume business. Considered a well-guarded secret by many, these fabs welcome low- and moderate-volume analogue business and offer pricing quite competitive with the billion-dollar fabs in Asia.

There is a time and place where standard analogue cells are more than adequate. Experienced analogue ASIC design houses recognise this and offer full-custom analogue designs only when the need merits it.

Myth #4. Using existing IP from analogue cell libraries lowers the chip cost.

FALSE. Using predesigned, functional cells such as amplifiers, con-